MODWT-Based Feature Extraction for OOK Demodulation in Visible Light Communication

Lorenzo Moreira Donatti
Federal University of Santa Maria
Santa Maria, Brazil
lorenzo.donatti@acad.ufsm.br

***Abstract*—Visible Light Communication (VLC) systems are emerging as a promising technology for high-speed wireless communication, particularly in environments where traditional radio frequency methods face limitations. However, VLC signals are highly susceptible to noise and signal degradation, especially as the distance from the transmitter increases. In this study, we propose a novel approach that combines Maximal Overlap Discrete Wavelet Transform (MODWT) with a Random Forest classifier to enhance the classification of On-Off Keying (OOK) modulated VLC signals. The MODWT is employed to extract robust features from the noisy signals, capturing critical temporal and frequency characteristics that improve classification accuracy. The Random Forest classifier then leverages these features to distinguish between different signal classes effectively. Through comprehensive experiments, we demonstrate that our approach significantly improves classification performance compared to baseline methods, making it a compelling solution for enhancing the reliability of VLC systems in real-world applications.**

***Index Terms*—MODWT, Wavelet Transform, Visible Light Communication, Machine Learning, Random Forests**

I. Introduction

Visible Light Communication (VLC) has emerged as a promising technology for high-speed wireless communication, utilizing the visible light spectrum to transmit data [1]. With the growing demand for higher data rates and the increasing congestion of traditional radio frequency (RF) bands, VLC offers a viable alternative due to its abundant bandwidth and inherent security advantages [2]. However, despite its potential, VLC systems are susceptible to noise and signal degradation, particularly at greater distances [3]. These challenges are exacerbated in On-Off Keying (OOK) modulation, where the presence of noise can significantly impair signal integrity, leading to errors in data transmission [4]. Traditional methods for mitigating these issues often fall short in addressing the complexity of the signal degradation observed in real-world VLC environments [5].

To address these challenges, we propose a novel approach that leverages Maximal Overlap Discrete Wavelet Transform (MODWT) [7] for feature extraction and Random Forest [8] for the demodulation of OOK modulated VLC signals. The MODWT is well-suited for capturing both temporal and frequency characteristics of the signals [9], providing a robust set of features that enhance the classifier's ability to distinguish between different signal classes, even in the presence of noise [10]. Furthermore, MODWT eliminates the dependency on sample size, unlike Discrete Wavelet Transform [11]. The Random Forest classifier is then employed to effectively utilize these features for accurate signal classification. We conduct a series of experiments to validate the performance of our approach, demonstrating its effectiveness in improving classification accuracy and signal robustness across various distances.

The remainder of this paper is organized as follows: Section 2 reviews the dataset used in this work. Section 3 details the methodology, including the MODWT and Random Forest techniques employed. Section 4 presents the experimental results and discussion. Finally, Section 5 concludes the paper.

II. Data

The data used in this study were obtained from [12]. The selection of this dataset is based on the fact that, according to the authors themselves, it is one of the first datasets involving visible light communication signal modulation available in the literature.

This dataset consists of received data from a physical visible light communication system in eight different modulations: OOK, QPSK, 4-PPM, 16-QAM, 32-QAM, 64-QAM, 128-QAM, and 256-QAM. In this work, we focus exclusively on the OOK modulation.

Furthermore, for each modulation type, the number of samples spans four different values: 10, 20, 40, and 80 samples per transmitted signal. The data is labelled according to the number of samples, meaning there are four distinct sets of results for each modulation. The data were collected at various distances, ranging from 0 cm to 140 cm from the transmitter, with all samples maintaining consistent labels. The data is normalized, with values ranging from 0 to 1.

In this study, we will focus only on OOK modulation with 40 and 80 samples, varying distances from 0 to 140 cm. Fig. 1 shows the difference between the signal 0 cm far from transmitter (a) and the signal 100 cm far from transmitter (b).

III. Proposed Method

In this chapter, we present a method for classifying signals in visible light communication (VLC) using the Maximum Overlap Discrete Wavelet Transform [7] for feature extraction and a Random Forest [8] as the classifier. This method leverages the capability of the MODWT to capture multi-scale features from signals and the robustness of Random Forest in classification tasks, especially in scenarios with high

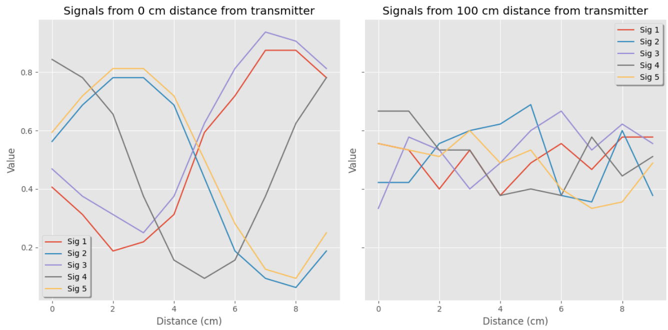

Fig. 1. Same signals in a) 0 cm, b) 100 cm distance from transmitter.

variability in the data [13]. The complete process of feature extraction and classification can be described as follows:

1) **Signal Preprocessing**: The input signals are preprocessed to remove potential distortions or anomalies that could affect the quality of the MODWT.
2) **MODWT Decomposition**: Each signal is decomposed using the MODWT into J levels, extracting detail coefficients $d_j[n]$ and approximation coefficients $a_j[n]$.
3) **Feature Extraction**: The features from the detail and approximation coefficients are extracted for each decomposition level, forming a feature vector $\mathbf{X}$.
4) **Classification**: The feature vector $\mathbf{X}$ is passed to the Random Forest classifier, which determines the signal's class based on the trained model.
5) **Evaluation**: To evaluate the performance of the method in visible light communication (VLC) signals, we used the classification accuracy.

A. *Maximum Overlap Discrete Wavelet Transform (MODWT)*

The Maximum Overlap Discrete Wavelet Transform (MODWT) is a variation of the Discrete Wavelet Transform (DWT) that eliminates the need for downsampling, maintaining translation invariance and temporal resolution of the signal [7]. This makes the MODWT particularly suitable for signal analysis where preserving temporal characteristics is essential.

Given an input signal $x[n]$, the MODWT decomposes the signal into different levels of detail and approximation. The decomposition is performed by applying high-pass filters $g^{(j)}[n]$ and low-pass filters $h^{(j)}[n]$ at each level j [14], as shown in the equations below:

$$d_j[n] = \sum_k x[k] \cdot g^{(j)}[n-k] \tag{1}$$

$$a_j[n] = \sum_k x[k] \cdot h^{(j)}[n-k] \tag{2}$$

where $d_j[n]$ and $a_j[n]$ represent the detail and approximation coefficients at level j, respectively. Since there is no downsampling, the length of $d_j[n]$ and $a_j[n]$ is the same as the original signal $x[n]$.

The features extracted from the MODWT coefficients at different decomposition levels are used to capture the multi-scale information of the signal. These features are then fed into a Random Forest classifier.

B. *Random Forest Classifier*

Random Forest is an ensemble of decision trees that improves classification accuracy by reducing overfitting and increasing model robustness [8]. Each decision tree in the ensemble is constructed from a random subset of the training set and attempts to minimize the classification error at each node using a criterion such as Gini impurity or entropy.

Given a set of features $\mathbf{X}$ extracted by the MODWT, the probability of a sample x_i belonging to a class c is obtained by aggregating the votes from all the trees in the forest, as expressed by the equation:

$$P(c \mid x_i) = \frac{1}{T} \sum_{t=1}^{T} P_t(c \mid x_i) \tag{3}$$

where T is the number of trees in the forest, and $P_t(c \mid x_i)$ is the probability assigned by the t-th tree.

The final class is then determined by the class that receives the most votes:

$$\hat{c} = \arg\max_c P(c \mid x_i) \tag{4}$$

C. *Architecture and Parameters*

For feature extraction, the MODWT has the following configuration:

- **Wavelet Type:** Daubechies 4 (db4)
- **Decomposition Level:** 3

The db4 wavelet, known for its compact support and efficient data representation, is particularly effective in preserving the signal's key features across different scales.

For classification, the Random Forest classifier is employed with the following parameters:

- **Number of Estimators**: 200
- **Criterion**: Entropy

The entropy criterion is chosen for its ability to handle complex classification tasks by measuring the information gain at each split in the decision trees.

IV. Results

In this chapter, we present the results obtained from applying the MODWT-based feature extraction and Random Forest classification method to visible light communication (VLC) signals. The primary objective is to evaluate the model's ability to classify signals with and without the wavelet transform across different sample sizes. The demodulation performance was evaluated using Accuracy, this metric was calculated for signals with 40 and 80 samples, both with and without the application of the MODWT, to measure the impact on classification.

Figure 2 display the demodulation performance metrics for 40 sample signals without and with the application of MODWT over distance. The results indicate a significant

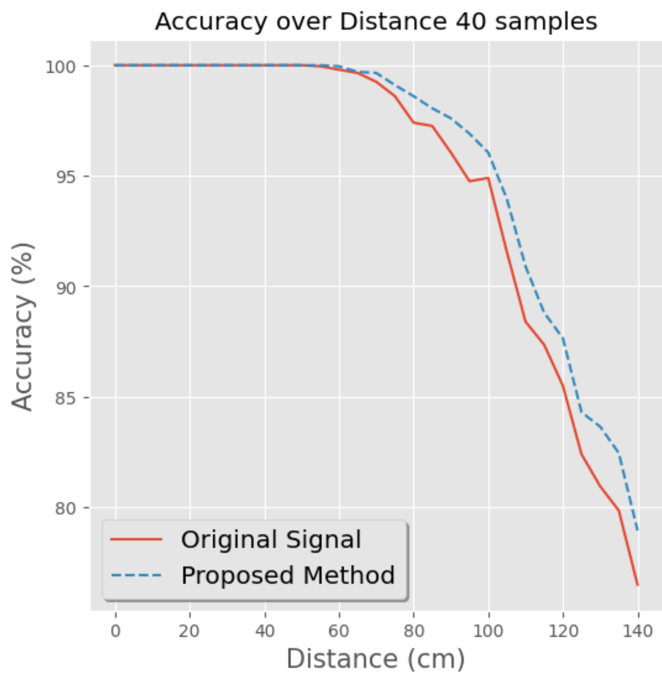

Fig. 2. Demodulation performance metrics for 40 sample signals over multiple distances.

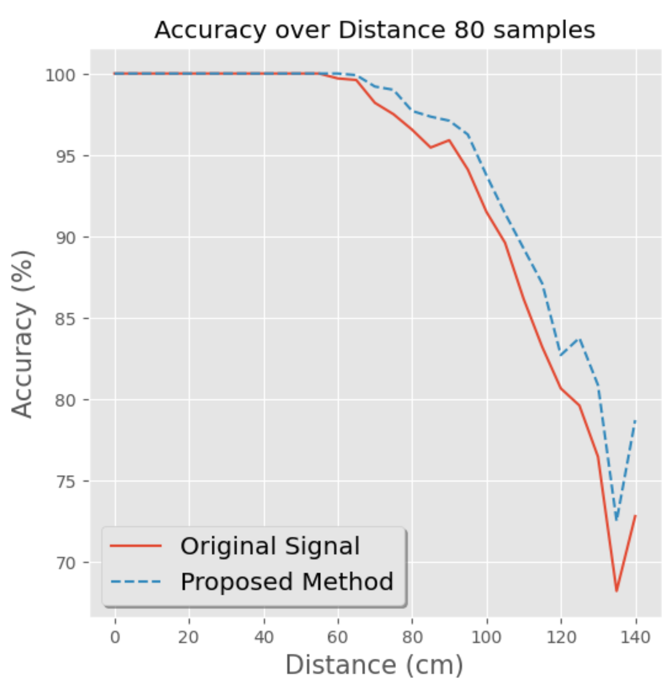

Fig. 3. Demodulation performance metrics for 80 sample signals over multiple distances.

improvement in classification accuracy after applying the MODWT in every distance.

Similarly, Figure 3 show the performance metrics for 80 sample signals. The application of MODWT again led to a notable enhancement in classification performance in all distances measured.

These results highlight the effectiveness of the MODWT in enhancing the feature representation of VLC signals, leading to improved demodulation performance. The Random Forest classifier, when combined with the wavelet-transformed features, consistently outperforms the baseline classification without wavelet preprocessing. This underscores the importance of wavelet-based methods in complex signal classification tasks.

V. Conclusion

In this study, we investigated the application of the Maximal Overlap Discrete Wavelet Transform (MODWT) combined with a Random Forest classifier for signal demodulation of visible light communication (VLC) signals. The primary objective was to enhance the classification accuracy by extracting robust features from the signals using the MODWT, which were then used to train the Random Forest model.

Our results demonstrate that the inclusion of MODWT for feature extraction significantly improves the classification performance compared to the baseline approach without wavelet transformation. This is evident in the higher accuracy metrics achieved across various sample and distance sizes. The wavelet-transformed features enabled the classifier to better capture the intricate patterns in the VLC signals, leading to more accurate and reliable classifications.

The findings of this study highlight the effectiveness of wavelet-based feature extraction in complex signal demodulation tasks, particularly in environments with noisy and degraded signals. By leveraging the strengths of MODWT in multi-resolution analysis and the robust classification capabilities of Random Forest, this approach shows great potential for improving signal processing and classification in VLC systems. Future work could explore the application of this method to other types of signals and further refine the feature extraction and classification process to achieve even higher performance.

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